Mobile Phone Feature Extraction and Price Prediction

Presented by: Mohit

Introduction

Mobile phones have different features, such as RAM, battery capacity, storage, camera quality, and processor performance. These features influence the price of a mobile phone.

This project uses Machine Learning to study mobile phone features and predict prices based on the available dataset.

Main objective: To understand which features influence mobile phone prices and build a model that can estimate prices.

Objectives

- To explore the mobile phone dataset.
- To clean and prepare the data for analysis.
- To identify important mobile phone features.
- To build a Machine Learning model for price prediction.
- To evaluate the model's performance.
- To understand how mobile phone features affect prices.

Dataset Description



The dataset contains information about mobile phones and their features.

Important features may include:

- RAM: Temporary memory used by the phone.
- Storage: Space available for apps, photos, and files.
- Battery: Battery capacity of the phone.
- Camera: Camera specifications.
- Processor: Component that processes instructions.
- Price (Prize): The target value that the model predicts.

Data Preprocessing

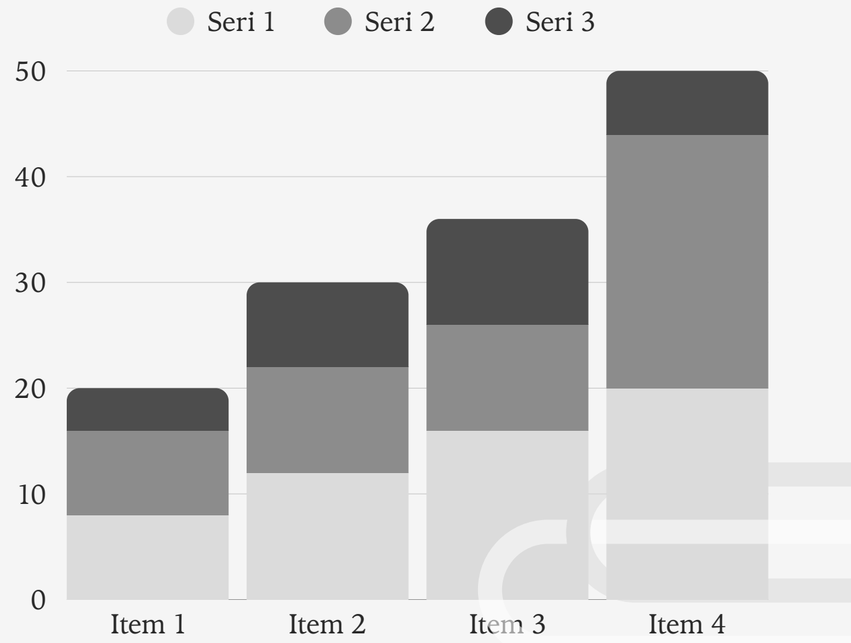
Data preprocessing means preparing raw data before applying Machine Learning.

The following steps are used:

1. Data Inspection: Check the rows, columns, and data types.
2. Missing Values: Identify and handle empty values.
3. Duplicate Removal: Remove repeated records.
4. Data Type Correction: Convert columns into appropriate data types.
5. Categorical Encoding: Convert text categories into a format suitable for the model.
6. Feature and Target Separation: Separate input features from the price column.

Why is preprocessing important?

It helps improve data quality and makes the dataset suitable for model training.



Exploratory Data Analysis (EDA)

EDA helps us understand patterns and relationships in the dataset through statistics and graphs.
The project uses:

- Histogram: To understand the distribution of mobile phone prices.
- Scatter Plot: To examine the relationship between RAM and price.
- Scatter Plot: To examine the relationship between battery capacity and price.
- Summary Statistics: To understand the dataset's numerical values.

Purpose: To discover patterns that may help explain differences in mobile phone prices.

Machine Learning Model

Model Building

The project uses the Random Forest Regressor algorithm to predict mobile phone prices.

How does Random Forest work?

- It builds multiple decision trees.
- Each tree makes a prediction.
- The predictions are combined to produce the final result.
- Steps followed:
- Split the dataset into training and testing sets.
- Train the model using the training data.
- Predict prices for the test data.
- Compare predicted prices with actual prices.
- The dataset is split into 80% training data and 20% testing data in the current code.

Conclusion

This project demonstrates how Machine Learning can be used to predict mobile phone prices from product features.

The project involves data exploration, data preprocessing, visualization, model building, and performance evaluation.

The Random Forest Regressor provides a method for estimating prices from the available data. Feature importance analysis can also help identify which features are most useful for prediction.

Final takeaway: Machine Learning can support data-driven pricing decisions, but the reliability of predictions depends on the quality of the dataset and the model's measured performance.

Thank You